Planetary Defense (grade 6 edition) — play-through review

Theme `planetary_defense_ms` · middle-school space science, grade 6 · 9 days, 33 stops · reviewed 2026-08-21 by reading `books/planetary-defense-ms.yml` in full, working every board, and comparing against the parent `planetary_defense`.

Verdict

The junior edition with the most ambitious arithmetic, and it mostly earns it. Eight boards, all correct:

Board	Arithmetic	Idea
How fast is it moving?	12 ÷ 20 = 0.6 arcsec/min, so ~36 an hour	an angular rate
Same brightness, different size	a quarter of the area, so half the width	area against length
How far away, from the echo?	4 s × 300,000 km/s = 1,200,000; half is 600,000	radar ranging
How much energy does it carry?	1,600 shared at 4 per megaton ≈ 400 megatons	energy in familiar units
Twice as fast	$0.5 \times 7.9 \times 10^9 \times (12^2 + 11.2^2) \times 10^6 \approx 1.06 \times 10^{18}$ J	the square on speed
Thirty-seven paths in a hundred thousand	37 in 100,000 ≈ a third of one in a thousand	a probability restated
How much does the push move it?	475 km/yr × 8 years ≈ 3,800 km	an early push does more
How many people is that share?	9,000,000 × 0.005 = 45,000	a percentage of a population

Two are unusually good for the age. **"Same brightness, different size"** — a quarter of the area means half the width — is the inverse-square-and-albedo degeneracy reduced to one sentence a child can hold, and it is the campaign's central ambiguity ("nobody knows yet whether this rock is the width of a street or of a town"). And **"Thirty-seven paths in a hundred thousand"** restates a probability rather than computing one, which is the right move: the difficulty at grade 6 is not the division, it is knowing that 37 in 100,000 is small *and not zero*.

Answerable: 33/33. **Sense:** Excellent. Four hours of observation, some paths going through the Earth, and a radar window that comes once in eleven years. **Level:** Right. `questionLoad` reports 4 of 33 demanding stops (12%). **Fun:** High. The parent's night-time ridge is the best-looking place in the junior set.

Implemented since this review

- **PM-05**, the opening card now names Dr. Anna Fischer.

Everything else in the findings table below is unchanged, and `npm run check` is green on this theme.

Findings

> ***Closed on 2026-08-22.*** *The rows marked CLOSED below were fixed and verified after the first pass; see* `_rejudged.pdf` *for the re-read.*

ID	Sev	Where	Issue	Proposed fix
PM-01	CLOSED	The syllabus: `momentum = weight × speed`, `change in speed = momentum ÷ weight`	**Two syllabus equations use weight where mass belongs, and one of them is the campaign's headline calculation.** Momentum is mass × velocity; weight is a force in newtons. A sixth grader who learns "momentum = weight × speed" has to unlearn it, and the deflection thread — which is the whole campaign — rests on it. The impact-energy board's own prompt reads "*energy = ½ × weight × speed × speed*", which is the same substitution in the formula a physics teacher fights hardest over. Note the inconsistency inside one syllabus: the sibling equation is written `KE = ½ × m × v²`, in symbols, with `m` — so the register is not even consistent about it.	Use "how heavy it is", which reads at grade 6 and is not wrong: *momentum = how heavy it is × how fast it is going*, *change in speed = momentum ÷ how heavy it is*, and *energy = ½ × how heavy it is × speed × speed*. Then make `KE = ½ × m × v²` match the register instead of standing out in symbols. Four other junior syllabi carry the same substitution — cross-campaign §11.
PM-02	PLAN	Seven of nine days author 4 stops	`per-day` is `[4,4,3,3,4,4,4,4,3]`. **Seven of nine days at the loop's maximum is the highest density in the catalogue**, and it means seven of nine days carry no callback. A nine-day campaign with 33 stops is 3.7 a day, so this is a length problem rather than an authoring accident: the content wants ten or eleven days.	**Reclassified, and the reasoning corrected.** I wrote that seven four-stop days are "why this edition has no callback". That is only half true: a callback also needs an unserved `— Review` variant, and this edition authors none — so it would have no callback at three stops a day either. The day load is still worth fixing and it is now one half of a two-part plan; see `_plan.pdf`.
PM-03	**WORTH**	`engine/dev/curriculum-debt.json`, 5 rows	Five of eight syllabus relations recorded as uncomputed: `average = total ÷ how many`, `KE = ½ × m × v²`, `momentum = weight × speed`, `change in speed = momentum ÷ weight`, `change in`	Re-check each row against the boards before treating it as content work, then fix the matcher (BM-01a)

			`position = change in speed × time`. **At least two look false** — "Twice as fast" computes $\frac{1}{2}mv^2$ explicitly and "How much does the push move it?" computes a change in position from a rate and a time. Same prose-and-symbol matcher problem as `blackout_ms`'s BM-01, compounded here because this syllabus mixes symbols and prose in one list.	rather than writing stops that already exist. `average = total ÷ how many` looks like the one genuine gap, and a campaign whose whole subject is *more observation separates two candidate paths* has an obvious home for it.
PM-04	**WORTH**	`engine/dev/concept-debt.json`, 5 rows	Five ordering rows, no `takesAsRead` at grade 6. CLAUDE.md records that moving the risk-cloud stop off day 2 cleared a row and put it after the concept it rests on — so this campaign has already been re-ordered once and these five are the residue. Two point at *impact energy: mass and speed, and why speed matters more*, which the "Twice as fast" board teaches.	If PM-02 adds a day, do the concept work in the same pass — the re-day is a chance to place the five bases before their dependents at no extra cost.
PM-05	**WORTH**	`opening:`	The card names nobody. It is otherwise the sharpest junior opening in the set — "*Nobody knows yet whether this rock is the width of a street or of a town. That is the difference between one city and one country*" — and the person missing is whoever is arguing for the radar window against whoever wants a number today.	One clause. Cross-campaign §8.
PM-06	**WORTH**	Zero `— Review` variants	No spaced retrieval. Cross-campaign §9. Worth more here than in most: this campaign's ideas (albedo against size, a probability that is small and not zero, an early push doing more) are exactly the kind a child needs to meet twice.	Three or four variants, best added in the same pass as PM-02.

What the derivation did well

- **It kept the ambiguity.** A rock the width of a street or of a town, and the campaign refuses to resolve it early.
- **The radar window.** One chance in eleven years is a deadline a child understands, and the ending says the window "is what proved it had worked".
- **"None of that quiet was luck."** The closing paragraph's first line, and the best statement in the junior set of why a right decision looks like nothing happening.

Opening and closing

Opening: 96 words. All four beats except the person (PM-05).

Closing: three paragraphs. "Nobody evacuated Valle Seco. The school there reopened on the Monday" is the right way to state a non-event as an achievement.

Warm-ups

Authored for this edition. This is one of the four two-tier junior sites, so it carries `trial-far` as well. No findings.